

Azure Virtual Machines

VM size families, disk types, availability options, extensions, and connecting securely — everything you need to deploy and reason about an Azure VM on the exam.

What a VM actually is in Azure

An Azure VM is a compute resource running on shared physical infrastructure, isolated by a hypervisor. You choose the OS image, size (CPU/RAM), disk configuration, virtual network, and authentication method. Azure manages the physical host — you manage everything inside the OS boundary. Understanding what you own vs. what Azure owns is the core of every VM-related exam question.

VM size families

B-SERIES

Burstable

Accumulates CPU credits when idle, spends them during bursts. Cheapest option for low-average-load workloads.

D-SERIES

General Purpose

Balanced CPU-to-memory ratio. Default choice for most workloads: web servers, app servers, dev/test.

E-SERIES

Memory Optimized

High memory-to-CPU ratio. Best for in-memory databases, large caches, SAP HANA.

F-SERIES

Compute Optimized

High CPU-to-memory ratio. Best for batch processing, game servers, analytics.

N-SERIES

GPU

GPU-equipped. Best for ML training, rendering, visualization, and CUDA workloads.

L-SERIES

Storage Optimized

High disk throughput and I/O. Best for NoSQL databases, data warehousing, large transactional DBs.

Availability options

SINGLE VM

No redundancy

99.9% SLA with Premium SSD. Vulnerable to host hardware failure. Not for production.

AVAILABILITY SET

Fault + Update Domains

Distributes VMs across fault domains (separate racks/power) and update domains (separate patch cycles). 99.95% SLA. Same datacenter.

AVAILABILITY ZONE

Zone-level redundancy

Distributes VMs across physically separate datacenters in one region. 99.99% SLA. Survives datacenter failure.

VMSS

VM Scale Set

Auto-scales a set of identical VMs based on demand or a schedule. Can span zones. Ideal for stateless web tiers.

SLA ladder: Single VM (Premium SSD) 99.9% → Availability Set 99.95% → Availability Zone 99.99%

Eight terms you need cold

OS Disk

The managed disk containing the operating system. Automatically created with the VM. Persists when the VM is stopped/deallocated.

Tip: Use Premium SSD for the OS disk on production VMs — it also unlocks the 99.9% single-VM SLA.

Managed Disk

Azure-managed storage for VM disks. Four tiers: Ultra Disk, Premium SSD v2, Premium SSD, Standard SSD, Standard HDD.

Tip: Always use Managed Disks — unmanaged disks are legacy and no longer recommended.

Temp Disk

A local SSD attached to the physical host — not persisted across deallocate or host migration. Used for swap / page file only.

Tip: Never store data on the temp disk — it is wiped on every reallocation.

VM Extension

A small agent that runs post-deployment inside the VM — Custom Script, DSC, MMA, AADSSHLogin, etc. Used to configure VMs without manual SSH/RDP.

Tip: Extensions are idempotent — re-running them applies config, not duplicates it.

Generalize / Sysprep

Removes machine-specific info (SID, hostname, user accounts) from a Windows VM before capturing it as a reusable image. Linux equivalent is `waagent -deprovision`.

Tip: A generalized VM cannot be started again — capture the image immediately after.

Azure Compute Gallery

A repository for storing, versioning, and replicating custom VM images and applications across regions and subscriptions.

Tip: Replaces the older "Managed Image" approach — ACG supports image versioning and cross-region replication.

Bastion

A PaaS service that provides browser-based RDP/SSH to VMs directly in the Azure Portal, over TLS 443 — no public IP or open port 22/3389 required on the VM.

Tip: The recommended way to connect to VMs in production — no jump box, no public IP exposure.

JIT VM Access

A Microsoft Defender for Cloud feature that opens RDP/SSH ports only for the duration of an approved request, then auto-closes them.

Tip: Requires Microsoft Defender for Cloud on the subscription — adds time-bound NSG rules automatically.

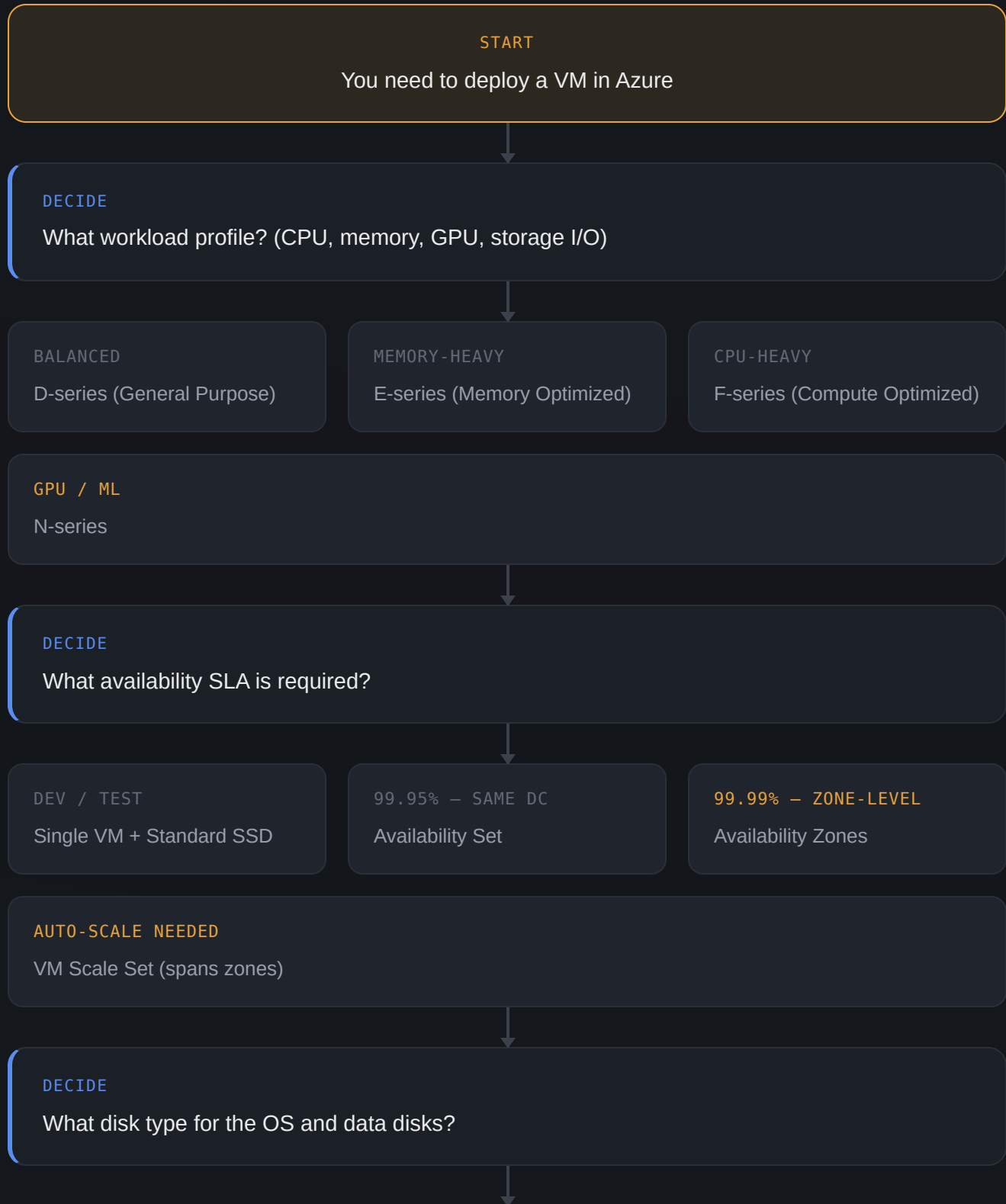
DISK TYPE	USE CASE	MAX IOPS (APPROX)
Ultra Disk	Latency-sensitive databases (SAP, SQL)	Up to 160,000
Premium SSD v2	High-perf production workloads	Up to 80,000
Premium SSD	Production VMs, I/O-intensive apps	Up to 20,000
Standard SSD	Web servers, dev/test	Up to 6,000
Standard HDD	Backups, non-critical workloads	Up to 2,000

EXAM GOTCHA – STOP VS DEALLOCATE

Stopping a VM from inside the OS does NOT release the compute allocation — you still get billed for compute. **Deallocating** (Stop in portal / `az vm deallocate`) releases the compute and stops billing, but the VM loses its dynamic public IP and may move to a different host on restart.

Deploying a VM — decision by decision

Every VM deployment comes down to the same set of choices. Walk through these in order.



PRODUCTION

Premium SSD (unlocks 99.9% single VM SLA)

DEV / TEST

Standard SSD or Standard HDD

DECIDE

How will you connect to the VM securely?

NO PUBLIC IP NEEDED

Azure Bastion (RDP/SSH via browser over TLS 443)

PORT-LEVEL JIT

JIT VM Access via Defender for Cloud

INTERNAL ONLY

Private IP + VPN / ExpressRoute

DECIDE

Does post-deployment configuration need to be automated?

YES

VM Extensions (Custom Script, DSC) or cloud-init (Linux)

NO

Manual or image-baked config

DONE

VM deployed with the right size, availability, storage, and access model

Tools for deploying and managing VMs

Every operation from creation to resizing to extension deployment can be driven from any of these tools.

Azure Portal

BEST FOR: first deploys, visual sizing, Bastion setup

The VM creation wizard walks through every decision — size, availability, disks, networking — and generates the ARM template at the end if you need it for IaC.

```
Virtual Machines → + Create → Azure
virtual machine
  → Review configuration → Download
ARM template
```

Azure CLI

BEST FOR: scripted deploys, pipelines, resizing

Fast VM creation and lifecycle management in single commands — easy to parameterize for multiple environments.

```
az vm create \
  --resource-group "rg-prod" \
  --name "vm-web01" \
  --image "Ubuntu2204" \
  --size "Standard_D2s_v3" \
  --admin-username "azureuser" \
  --generate-ssh-keys

az vm resize \
  --resource-group "rg-prod" \
  --name "vm-web01" \
  --size "Standard_D4s_v3"

az vm deallocate \
  --resource-group "rg-prod" --name
"vm-web01"
```

> Azure PowerShell

BEST FOR: Windows-centric / enterprise automation

Full feature parity with CLI — preferred by teams already invested in PowerShell for Windows workloads.

```
New-AzVM -ResourceGroupName "rg-prod" `
  -Name "vm-web01" -Location "eastus" `
  -Image "UbuntuLTS" -Size "Standard_D2s_v3" `
  -Credential (Get-Credential)

Stop-AzVM -ResourceGroupName "rg-prod" `
  -Name "vm-web01" -Force
```

<> Bicep / ARM

BEST FOR: IaC, repeatable environments

Define VM, NIC, disk, NSG, and public IP as a single deployable unit — version-controlled and idempotent.

```
resource vm
'Microsoft.Compute/virtualMachines@2023-03-01' = {
  name: 'vm-web01'
  location: location
  properties: {
    hardwareProfile: { vmSize:
'Standard_D2s_v3' }
    osProfile: {
      computerName: 'vm-web01'
      adminUsername: adminUsername
    }
    storageProfile: {
      imageReference: {
        publisher: 'Canonical'
        offer: 'UbuntuServer'
        sku: '22_04-lts'
        version: 'latest'
      }
    }
    networkProfile: {
      networkInterfaces: [{ id:
nic.id }]
    }
  }
}
```


Run Command

BEST FOR: emergency access, config without RDP/SSH

Execute scripts inside a VM directly from the portal or CLI when you can't connect via RDP/SSH — useful for enabling Remote Desktop, resetting passwords, or diagnosing issues.

```
az vm run-command invoke \  
  --resource-group "rg-prod" \  
  --name "vm-web01" \  
  --command-id RunShellScript \  
  --scripts "sudo systemctl restart  
nginx"
```

TOOL	BEST FOR	SUPPORTS EXTENSIONS?	REPEATABLE?
Portal	First deploy, visual config	Yes	No
CLI	Scripting, pipelines	Yes	Yes
PowerShell	Windows enterprise automation	Yes	Yes
Bicep / ARM	IaC, environments as code	Yes	Yes
Run Command	Emergency in-VM execution	No	No

Principles worth internalizing

OPTION	PROTECTS AGAINST	SLA	COST IMPACT
Single VM + Premium SSD	Routine maintenance only	99.9%	Baseline
Availability Set	Rack/power failures + patching waves	99.95%	Low — same DC
Availability Zone	Full datacenter failure	99.99%	Moderate — zone egress
VMSS (multi-zone)	Datacenter failure + demand spikes	99.99%	Scales with load

EXAM GOTCHA — AVAILABILITY SETS VS ZONES

Availability Sets operate within a single datacenter — they protect against rack failures and staggered updates, but a full datacenter outage takes out all VMs. **Availability Zones** put VMs in separate physical datacenters within a region — if one zone loses power, the others keep running. You cannot move an existing VM into an Availability Set or Zone after creation.

CONNECTING SECURELY — HIERARCHY

Best: Azure Bastion (no public IP, no open ports, browser-based, TLS 443). **Good:** JIT VM Access (port open only during approved window). **Avoid in production:** static public IP with port 22 or 3389 open — exposed to brute-force constantly.

Six mistakes that show up on the exam

Stopping a VM from the OS and expecting no billing

OS-level shutdown keeps the VM allocated — you're still billed. Use Deallocate (portal Stop or `az vm deallocate`) to stop compute billing.

Storing data on the temp disk

The temp disk (`D:` on Windows, `/dev/sdb` on Linux) is wiped on every reallocation, host migration, and resize — it's for swap/page file only.

Using Standard HDD and expecting 99.9% SLA

The single-VM 99.9% SLA only applies when the OS disk is Premium SSD (or better). Standard disks drop you below that threshold.

Trying to add a VM to an Availability Set after creation

You cannot change the availability configuration of a deployed VM — it must be set at creation time.

Generalizing a VM and then trying to start it

After Sysprep / waagent deprovision, the VM is non-bootable — it must be captured as an image immediately.

Confusing VM Extensions with VM Applications

Extensions run management agents post-deploy (DSC, monitoring). VM Applications install software packages from a Compute Gallery — different mechanism, different use case.